Company. Designed during World War I, this drone was intended for defensive experimentation as part of Britain's early efforts to create unmanned interceptors. The aircraft featured a lightweight wooden biplane structure with fabric-covered wings, a small piston engine, and a propeller-driven design. Its flight control was achieved through radio-transmitted signals that activated servomechanisms linked to the rudder and elevators. Though limited by the primitive radio and power technology of its time, the Ruston Proctor Aerial Target became one of the first successful demonstrations of powered unmanned flight, validating the feasibility of radio-guided control and laying the groundwork for future developments in UAV systems.



Figure 2.7 The Ruston Proctor Aerial Target (1917) – early pilotless flight experiment.⏎

2.7 Limitations of First-generation Drones

Despite their historical importance and pioneering role in the evolution of unmanned aerial vehicles (UAVs), first-generation drones were constrained by numerous technological, operational, and practical limitations. These early systems represented the infancy of drone innovation relying on